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AUTOMATED IDENTIFICATION OF MEDICAL PRESCRIPTION

Thesis submitted in partial fulfillment of curriculum prescribed for the
award of the degree of

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND BUSINESS SYSTEMS

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July, 2023

DECLARATION

- I. We certify that the work contained in this report has been done by us under the guidance of our supervisor Dr. B S Mahanand, Head of the Department, Department of Information Science and Engineering, JSS Science and Technology University, Sri Jayachamarajendra College of Engineering.
- II. The work has not been submitted to any other Institute for any degree or diploma.
- III. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- IV. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.
- V. The dissertation work is submitted in partial fulfillment of the requirements for the award of degree of Bachelor of Engineering in Information Science and Engineering for the academic year 2019 – 2023.

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ABSTRACT

This report describes the finds and results of a project aimed at creating an automated system for identifying and deciphering prescriptions. The project addresses the growing demand for precise and efficient prescription filling in healthcare facilities. The automated identification system seeks to improve patient safety by streamlining prescription management, reducing human mistakes and utilizing advances in machine learning and natural language processing techniques to efficiently identify the drugs in medical prescriptions.

The report begins with an introduction on current difficulties with manually processing prescriptions, including the possibility of mistakes, delays and misinterpretation. It underlines the need of automating this procedure to boost effectiveness and keep the patient informed about the drugs being consumed and by also suggesting alternatives. Next, the report describes the methodology adopted to develop the automated identification system. This involved a multi-step approach that included data collection, preprocessing, feature extraction, and model training. The system was trained on three different datasets.

Further, the findings and the performance of all the models on various datasets are discussed. We have also curated a custom data set using IoB tagging which is further explained in the report.

Furthermore, the report describes the potential applications and benefits of implementing the automated identification system in real-world healthcare settings. It highlights the system's potential to improve prescription accuracy, reduce human errors, enhance patient safety and expedite prescription processing.

In conclusion, this report underscores the importance of automated identification systems in medical prescription processing. The project's findings demonstrate the feasibility and effectiveness of employing advanced machine learning techniques. This project holds immense promise in revolutionizing and benefiting healthcare professionals and patients alike.

ACKNOWLEDGEMENT

First and foremost, we are indebted to our college JSS Science and Technology University, Mysuru for providing us all the facilities needed for the successful completion of our project work.

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this final year project.

We are deeply thankful to our project guide (supervisor), Dr. B S Mahanand, for his invaluable guidance, expertise, and continuous support throughout the entire duration of this project. His instructive feedback, constructive criticism and encouragement have been instrumental in shaping the direction and quality of this work.

We would also like to extend our appreciation to the faculty members of JSS Science and Technology University, for their valuable inputs, suggestions, and encouragement during the course of this project. Their expertise and knowledge have significantly enriched my skill development in the field.

To all those mentioned above and to anyone else who has directly or indirectly contributed to this project, we extend our heartfelt appreciation. Your support and assistance have been invaluable and this project would not have been possible without you.

Thank you

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INTRODUCTION

1.1 INTRODUCTION

The automation of healthcare processes has become increasingly essential in recent years, driven by technological advancements and the need for improved efficiency and patient safety. One critical area that can benefit from automation is the identification of medical prescriptions. The accurate identification of prescriptions is crucial for ensuring patient safety, minimizing errors, and optimizing healthcare workflows. Manual identification of prescriptions is time-consuming, error-prone, and often subject to variations in handwriting styles and prescription formats. Therefore, developing an automated system for prescription identification holds significant promise for enhancing healthcare practices.

The objective of this project is to design and develop an automated identification system for medical prescriptions. The system aims to accurately extract relevant information from prescriptions, including medication names, dosages, instructions, and patient details. By automating this process, the project seeks to improve efficiency, reduce errors, and enhance patient safety in healthcare settings. Additionally, the project aims to standardize prescription identification practices and facilitate seamless integration with existing electronic health record (EHR) systems, enabling comprehensive patient care and data management.

The potential benefits of automating prescription identification are far-reaching. By reducing the time and effort required for manual identification, healthcare providers can dedicate more time to patient care and critical tasks, leading to improved workflow efficiency and productivity. Moreover, automation can help minimize medication errors, ensuring that patients receive the correct medications and dosages, thus enhancing patient safety and adherence.

This project aims to contribute to the broader field of healthcare automation by addressing the specific challenges and requirements of automated prescription identification. By developing a reliable and accurate system, the project seeks to improve healthcare practices, standardize identification processes, and facilitate seamless integration with existing healthcare systems.

In the subsequent sections of this report, we will delve into the techniques, methodologies, and considerations involved in developing the automated prescription identification system. We will explore the data collection and annotation process, the training and fine-tuning of models, evaluation metrics, and the application of the system in real-world healthcare scenarios. Additionally, we will discuss the challenges faced during the project, ethical considerations, and potential future directions for automated prescription identification.

By undertaking this project, we aim to contribute to the advancement of healthcare technology, improve patient care and safety, and pave the way for more efficient and accurate prescription identification processes in healthcare settings.

1.2 ABOUT THE PROJECT

1.2.1 Motivation

With a rapid advancement of technology, there is a growing need for automation in various industries, including healthcare. Automating the identification of medical prescriptions can streamline processes, improve accuracy, and enhance patient safety. Manual identification of medical prescriptions is prone to errors. By automating the identification process, the potential of errors can be significantly reduced, leading to improved patient care and outcomes. Automating prescription identification can contribute to better management of medication records and patient histories. By accurately capturing and organizing prescription data, healthcare providers can easily retrieve and analyze information, leading to more informed decision-making and improved patient care.

Automated prescription identification systems can serve as a valuable tool to support healthcare professionals by providing accurate and reliable information about prescriptions including drug names, dosages and administration instructions. These systems can assist healthcare providers in making informed decisions and reducing the likelihood of errors. Automation in prescription identification offers the opportunity for scalability and standardization across healthcare facilities. By implementing a consistent and reliable system, the challenges associated with deciphering various handwriting styles or prescription formats can be overcome ensuring accuracy and efficiency. Auto-mated prescription identification can seamlessly integrate with electronic health record systems (EHR), enabling efficient storage and retrieval of

prescription information. Recent technological advancements in natural language processing (NLP), machine learning and computer vision techniques have significantly improved the accuracy and reliability of automated identification systems. Leveraging these technological advancements can enable the development of robust and effective prescription identification solutions.

The field of automated prescription identification is still evolving, offering ample opportunities for research and development. Engaging in such a project can contribute to the advancement of healthcare technology, potentially leading to new insights, methodologies and solutions that benefit the healthcare industry as a whole. By considering these motivations, we developed a compelling rationale for undertaking a project on the automated identification of medical prescriptions.

1.2.2 Problem Definition

The problem that the project aims to address is the inefficiency, errors and potential risks associated with manual prescription identification processes in healthcare settings. The project seeks to develop a reliable and accurate automated system that can effectively identify and extract relevant information from medical prescriptions including medication names, dosage, instructions, and patient details. Key aspects of the problem definition include:

- a. Inefficiency and Time constraints - Manual identification of medical prescriptions is a time-consuming process, requiring healthcare professionals to review and decipher handwritten or complex prescriptions. This can lead to delays in patient care and hinder healthcare providers' productivity.
- b. Error-Prone Identification: Manual prescription identification is prone to errors, such as misinterpretation of handwriting or unclear instructions. These errors can result in medication mistakes, potentially compromising patient safety and leading to adverse drug reactions.
- c. Lack of Standardization: Medical prescriptions can vary in format, language and writing style, making the identification process challenging and prone to inconsistencies. This lack of standardization further exacerbates the risk of errors and hinders efficiency prescription management.
- d. Patient safety and adherence: Errors in prescription identification can have serious implications for patient safety, including incorrect medication

administration or dosage. Automating the identification process can help reduce these risks, ensuring patients receive the correct medications and dosages.

- e. **Complex prescription structures:** Medical prescriptions can contain complex structures, including abbreviations, dosage instructions and medication frequency. Manual identification of these structures can be challenging, requiring domain expertise and increasing the likelihood of errors.
- f. **Scalability and Consistency:** In larger healthcare settings or across multiple healthcare facilities, maintaining consistent and accurate prescription identification becomes increasingly difficult. Automating the identification process can provide scalability and standardization, ensuring consistent and reliable identification regardless of the volume or variety of prescriptions.

By defining these problem areas, the project aims to develop a solution that overcomes the limitations of manual prescription identification, enhances efficiency, reduces errors, improves patient safety, and integrates seamlessly with existing healthcare systems.

1.2.3 Challenges

- a. **Complex Prescription Formats:** Medical prescriptions can have complex structures and formats, including abbreviations, symbols, dosage instructions and frequency details. Deciphering and extracting the relevant information accurately from these complex structures pose a significant challenge for automated systems.
- b. **Handling Prescription Abbreviations:** Medical prescriptions often utilize various abbreviations specific to the healthcare domain. These abbreviations can be ambiguous and context-dependent, requiring sophisticated algorithms to accurately interpret their intended meaning and context.
- c. **Limited Training Data and Domain-Specific Challenges:** Obtaining a sufficient amount of accurately annotated training data for prescription identification can be challenging, especially for specialized domains or rare medical conditions. Building robust and accurate models may require extensive data collection efforts and expertise in domain-specific challenges.
- d. **Ongoing Model Updates and Maintenance:** Medical knowledge, drug information, and prescription practices continuously evolve. Therefore, maintaining and updating the automated identification systems to keep up with

new modifications, dosage guidelines, and emerging practices is essential. Regular model updates and system maintenance should be planned to ensure the system remains accurate and up-to-date.

- e. Indian Dataset for Medicine: Lack of an annotated dataset for Indian medicine names. The absence of localized data can raise concerns on the automated identification of local medicine prescriptions.

Addressing these challenges requires a combination of advanced machine learning and natural language processing techniques, domain expertise and extensive data collection. Overcoming these obstacles will lead to the development of robust and reliable automated identification systems for medical prescriptions, ultimately improving patient safety and health care efficiency.

1.2.4 Objectives

- **Develop an Accurate and Reliable Prescription Identification System:** The primary objective of the project is to design and develop an automated system that can accurately identify and extract relevant information from medical prescriptions. The system should demonstrate a high level of accuracy in deciphering prescription details, including medication names, dosages, instructions, and patient information.
- **Enhance Patient Safety and Reduce Medication Errors:** Another essential objective of the project is to contribute to patient safety by minimizing medication errors associated with manual prescription identification. By accurately identifying prescriptions and ensuring the correct medications and dosages, the system can help prevent medication mistakes, adverse drug reactions, and potential harm to patients.
- **Explore Potential for Scalability and Generalizability:** The project aims to explore the potential for scalability and generalizability of the automated prescription identification system. It seeks to assess the system's performance across different healthcare settings, including hospitals, clinics, and pharmacies, with diverse patient populations, prescription patterns, and languages.
- **Continuously Improve and Update the System:** An important objective is to establish a framework for continuous improvement and system updates. The project aims to monitor system performance, collect feedback from users, and

incorporate new data and knowledge to enhance the accuracy, reliability, and adaptability of the prescription identification system over time.

- **Ensure Compliance with Privacy and Ethical Guidelines:** The project aims to prioritize patient privacy, data security, and compliance with ethical guidelines throughout the development and implementation of the automated prescription identification system. It seeks to establish robust data governance practices, adhere to relevant regulations, and ensure the ethical use of patient information.

By addressing these objectives, the project aims to contribute to the advancement of healthcare technology, improve patient care and safety, enhance workflow efficiency, and establish standardized practices in the identification of medical prescriptions.

2.1 Medical Prescription Classification: An NLP Based Approach

Authors: Viincenza Carchiolo; Alessandro Longheu; Giuseppa Reitano; Luca Zagarella

Year Of Publication: 2019

The digitization of healthcare data has been consolidated in the last decade as a must to manage the vast amount of data generated by healthcare organizations. Carrying out this process effectively represents an enabling resource that will improve healthcare services provision, as well as on-the-edge related applications, ranging from clinical text mining to predictive modeling, survival analysis, patient similarity, genetic data analysis and many others. The application presented in this work concerns the digitization of medical prescriptions, both to provide authorization for healthcare services or to grant reimbursement for medical expenses. The proposed system first extract text from scanned medical prescription, then Natural Language Processing and machine learning techniques provide effective classification exploiting embedded terms and categories about patient/doctor personal data, symptoms, pathology, diagnosis and suggested treatments.

They have proposed the below system and the solutions to achieve the goal described above. First the input image is examined to establish the type and format of the medical prescription that image represents, then text is collected and corrected to further isolate and extract all relevant strings and the information based on previously collected strings is classified. Finally, the prescription is eventually considered as valid for further approval or not, according to specific criteria based on the information and related classification.

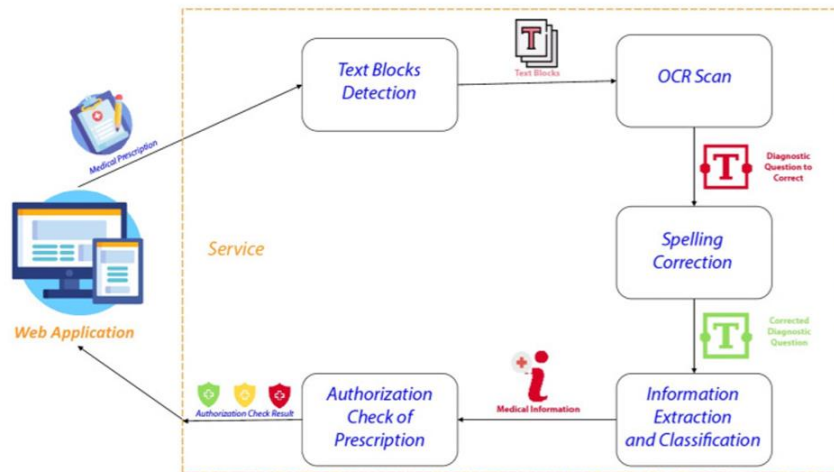


Figure 2.1.1

SERVIZIO SANITARIO NAZIONALE		RICETTA ELETTRONICA - PROMEMORIA PER L'ASSISTITO	
Regione Sicilia		[Barcode]	
COGNOME E NOME/INIZIALI DELL'ASSISTITO: [Redacted]			
INDIRIZZO: [Redacted]	CAP: 95100	CITTA': Catania	PROV: CT
INDIRIZZO: [Redacted]	STORIA PROVINCIA: CT	MODULO ADEL: 208	DISPOSIZIONI REGIONALI
TIPOLOGIA PRESCRIZIONE(S): [Redacted]	ALBOS: [Redacted]	PECORITA: [Redacted]	PRESCRIZIONE(U.B.D.P.): [Redacted]
PRESCRIZIONE		QTA	NOTA
90433 - TRIODOTIRONINA LIBERA (FT3)		[Redacted]	---
90423 - TIROXINA LIBERA (FT4)		[Redacted]	---
90421 - TIREOTROPINA (TSH)		[Redacted]	---
QUESTO DIAGNOSTICO: tireopatia - controllo			
RICETTA ELETTRONICA - PROMEMORIA PER L'ASSISTITO			
CODICE ASSISTENTE/CAPIFARMICO: 16152811117005070053303112222			

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    '90423 - tiroxina libera (ft4)',
    '90421 - tireotropina (tsh)'
  ],
  'quesito': 'tirepatia - controllo'
}
  
```

Figure 2.1.2

The first module of the proposed system aims to discriminate which type of medical prescription is being processed. After image pre-processing, text is extracted to build a string dictionary where each entry represents a field of the medical prescription.

Once the text has been extracted, we proceed with a spelling correction, that is required as in most OCR software residual errors still occur, in particular in our scenario where the quality of scanned images is not always high and also the test that appears in medical prescription is usually with a reduced font size.

The goal of information classification is to assess whether a given prescription is grantable or not, as specified in the paper. Syntactic rules and the Rule-based tagging NLP technique. Syntactic rules are used to model all valid grammar sequences, whereas Rule-based approaches use contextual information to assign tags to unknown or

ambiguous words (context former rules). Rule-based triggers generally require supervised training, but also other approaches are available.

Finally in the testing phase carried out we evaluate the performance of spelling correction and information classification algorithms, with the final purpose of establishing whether an input medical prescription can be classified as grantable or not, or eventually whether the system was not able to classify it at all.

The main goal of the proposed system is to develop a service for the analysis and authorization of medical prescriptions. Results showed that in most cases the system allows automatic classification of the text and the prescribed medicine and tests were correct with 5% error.

2.2 Efficient Estimation of Word Representations in Vector Space

Authors: Tomas Mikolov, Kai Chen, Greg Corrado, Jeffrey Dean

Year Of Publication: 2013

The paper presents a method for learning word representations, also known as word embeddings, in a high-dimensional vector space. The authors propose two neural network-based models, called Continuous Bag-of-Words (CBOW) and Skip-Gram, for learning these representations.

The CBOW model predicts a target word given its context, while the Skip-Gram model predicts the context given a target word. Both models are trained on a large corpus of text using a negative sampling technique, which is more computationally efficient than other methods such as hierarchical softmax. The authors of the paper evaluate the quality of the learned word representations on several natural language processing tasks, such as language modeling, part-of-speech tagging, and named entity recognition, and show that they outperform previous state-of-the-art models.

The paper also highlights the advantages of word embeddings, such as the ability to capture semantic and syntactic relationships between words, and their ability to be used in a wide range of NLP tasks. The method proposed in this paper has been widely used in many NLP applications and has become a standard in the field.

Word embeddings are a way of representing words as high-dimensional vectors in a continuous vector space. The goal of word embeddings is to capture the meaning of a word in a way that can be used as input to machine learning models, specifically neural networks. The idea behind word embeddings is to map words to a dense vector in a high-dimensional space, where semantically similar words are close to each other and dissimilar words are far apart.

For example, the word "cat" and "feline" might be close to each other in the vector space, while the word "cat" and "car" would be far apart. This is because they have similar semantic meanings, while the meaning of "cat" and "car" are very different.

Similarly, vector representations of words like "king", "queen", "prince" and "princess" would be close to each other in the vector space because they have similar semantic meaning, while the words like "king" and "lion" would be far apart.

Word embeddings can be used for a wide range of natural language processing tasks, such as language modeling, part-of-speech tagging, named entity recognition, sentiment analysis, and machine translation. They can also be used as input to other machine learning models, such as neural networks for text classification, question answering, and text generation.

Word embeddings are also useful for pre-training models for transfer learning in NLP tasks. It's a widely used technique in the NLP field, many pre-trained embeddings are available for use such as word2vec, GloVe, and BERT.

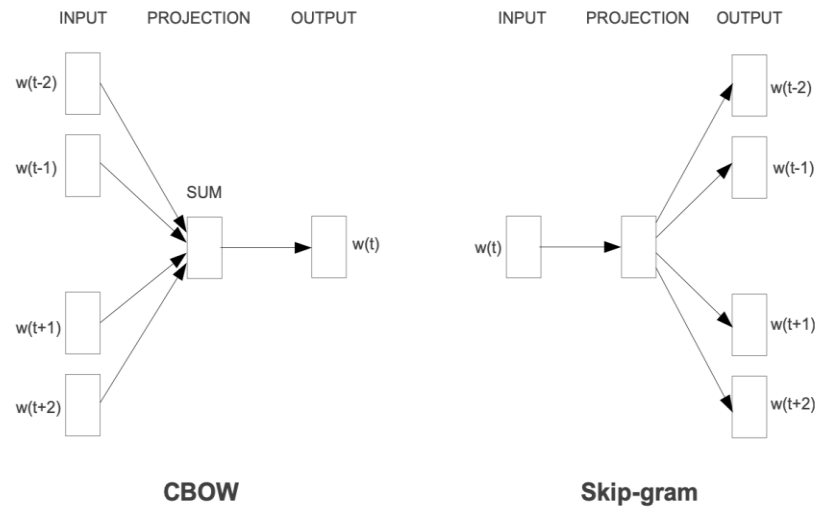


Figure 1: New model architectures. The CBOW architecture predicts the current word based on the context, and the Skip-gram predicts surrounding words given the current word.

Figure 2.2.1

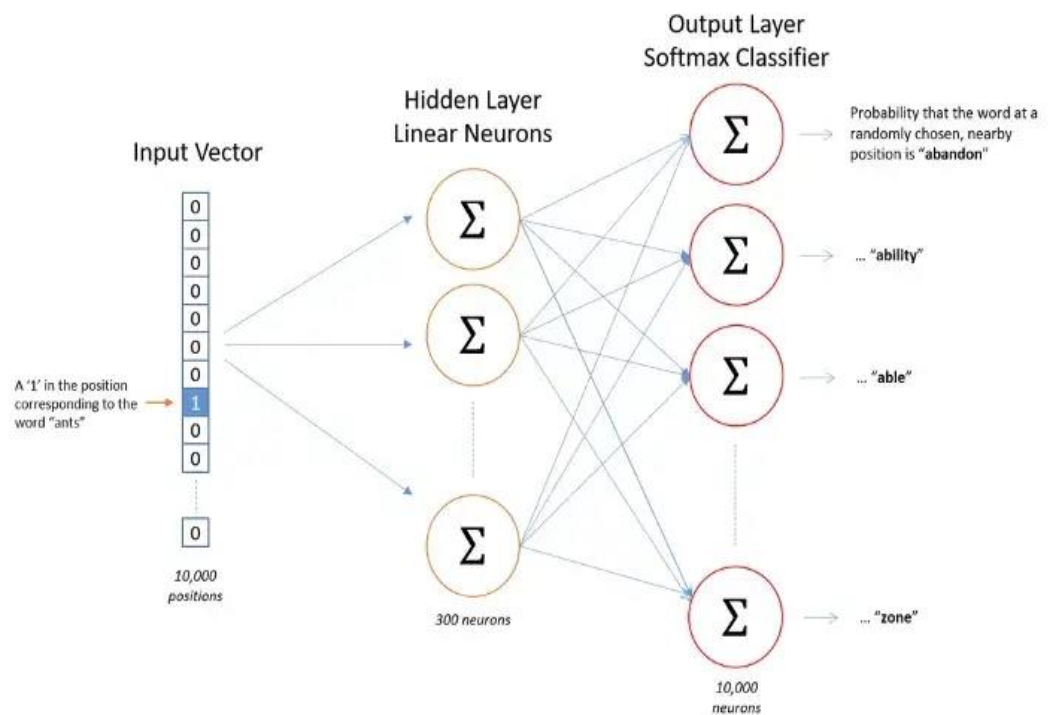


Figure 2.2.2

Table 3: Comparison of architectures using models trained on the same data, with 640-dimensional word vectors. The accuracies are reported on our Semantic-Syntactic Word Relationship test set, and on the syntactic relationship test set of [20]

Model Architecture	Semantic-Syntactic Word Relationship test set		MSR Word Relatedness Test Set [20]
	Semantic Accuracy [%]	Syntactic Accuracy [%]	
RNNLM	9	36	35
NNLM	23	53	47
CBOV	24	64	61
Skip-gram	55	59	56

Table 2.2.1

2.3 Recent Trends in Named Entity Recognition (NER)

Author: Arya Roy

Year Of Publication:2021

Named Entity Recognition (NER) is a sub-task of natural language processing (NLP) that aims to identify and classify named entities, such as people, organizations, locations, and dates, in text. The process of NER typically involves the following steps:

1. Tokenization: This is the process of breaking down the text into individual words or tokens. Tokens are the basic units that NER systems work with.
2. Part-of-speech (POS) tagging: This is the process of identifying the grammatical role of each token in a sentence. POS tags can help to disambiguate words with multiple meanings, such as "bank" (a financial institution or the edge of a river)
3. Chunking: This is the process of grouping words together into meaningful chunks, also known as chunks or noun phrases. This step can help to identify named entities that are composed of multiple words, such as "New York City"
4. Named Entity Recognition: This is the core step of the process, where the system identifies and classifies named entities. There are different methodologies used for this step, including rule-based systems, statistical models, and machine learning-based systems.
5. Evaluation: This is the step where the system's performance is evaluated on a set of labeled test data. The most common evaluation metric for NER is the F1 score, which is the harmonic mean of precision and recall.

There are several methods that can be used to implement Named Entity Recognition (NER), including:

1. Rule-based systems: These systems use a set of hand-crafted rules to identify and classify named entities. These rules are based on patterns in the text, such as capitalization, prefixes, and suffixes. Rule-based systems can be highly accurate, but they are limited by the coverage of the rules, and they are difficult to maintain and adapt to new domains and languages.
2. Statistical models: These models, such as Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs), use the statistical properties of the text to identify and classify named entities. These models can be trained on labeled data and they can take into account the context of the words in a sentence.
3. Machine learning-based systems: These systems use machine learning algorithms, such as Support Vector Machines (SVMs), decision trees, and neural networks, to learn the patterns in the text that are indicative of named entities. These systems can be trained on large amounts of labeled data, and they can achieve state-of-the-art performance on NER tasks.
4. Hybrid systems: These systems combine the strengths of different methods, such as using rule-based systems to pre-process the text and machine learning-based systems to classify named entities. These systems can be more robust and accurate than any of the individual methods.
5. Deep Learning-based systems: These systems use neural networks and their variants such as LSTM, GRUs, and transformer-based models such as BERT and its variants. These models are able to learn rich representations of words and context, which helps them to better identify and classify named entities.

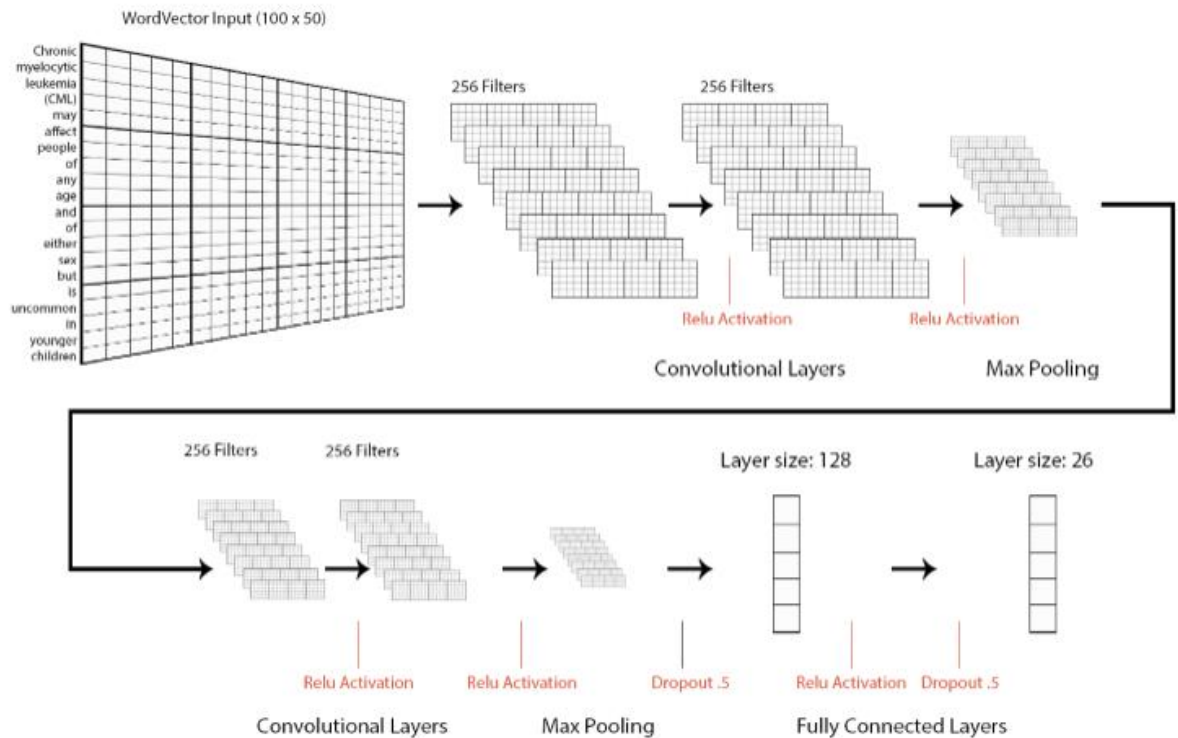


Figure 2.3.1

2.4 A Neural Attention Model for Sentence Summarization

Authors: Alexander M. Rush, Sumit Chopra, Jason Weston

Year Of Publication: 2015

This paper presents a neural attention-based model for the task of sentence summarization. The authors propose a combination of neural networks and attention mechanisms to extract the most important information from a text and reduce it to a shorter version that still conveys the main ideas.

Summarization is an important challenge of natural language understanding. The aim is to produce a condensed representation of an input text that captures the core meaning of the original. Most successful summarization systems utilize extractive approaches that crop out and stitch together portions of the text to produce a condensed version. In contrast, abstractive summarization attempts to produce a bottom-up summary, aspects of which may not appear as part of the original.

The model they proposed is based on an encoder-decoder architecture, where the encoder is used to embed the input sentence into a continuous representation, and the

decoder generates a summary by attending to the encoded input. In the attention mechanism, the model learns to focus on specific parts of the input sentence to generate a more informative summary.

The model's main components are:

1. **Encoder:** It encodes the input text into a fixed-length vector representation. It uses a bidirectional recurrent neural network (RNN) to process the input text, which allows the model to consider the context of the text in both the forward and backward directions. It takes the input text and converts it into a fixed-length vector representation.
2. **Attention Mechanism:** The attention mechanism used in the model is called "global attention", which assigns a weight to each word in the input text, indicating the importance of that word in the summary. The decoder then uses these weights to decide which words to include in the summary, and in what order to generate them.
3. **Decoder:** The decoder uses the encoded input text and the attention weights to generate the summary one word at a time. It uses the attention weights to decide which words to include in the summary and in what order to generate them.

The authors evaluated their model on several datasets, including the Gigaword and the DUC datasets, and showed that it outperforms the state-of-the-art models at the time of its publication. They also demonstrated the robustness of the model by testing it on a variety of input lengths and showed that it can handle both long and short sentences with high accuracy.

The attention mechanism is an important component of the model, which allows the model to learn to focus on specific parts of the input sentence to generate a more informative summary. The authors argue that this is a more powerful approach than traditional methods, which rely on heuristics or rule-based methods to select the most important sentences or phrases.

There are two main types of attention mechanisms: additive attention and multiplicative attention. Additive attention is also known as "soft attention" and involves computing a weighted sum of the input values, where the weights are learned by the model. This is done by computing a dot product of the input values and a set of learnable weights, and then applying a softmax function to the results to obtain the attention weights. The

final output is obtained by taking a weighted sum of the input values, where the weights are the attention weights.

Multiplicative attention, also known as "hard attention," involves computing an attention score between the input values and a set of learnable weights, which is then used to compute the attention weights. The attention weights are then multiplied with the input values to obtain the final output.

In practice, attention mechanisms are usually used in combination with other neural network architectures, such as recurrent neural networks (RNNs) or transformer networks, to allow the model to focus on specific parts of the input when processing it. Attention mechanisms can be used in a wide range of tasks, including natural language processing, computer vision, and speech recognition.

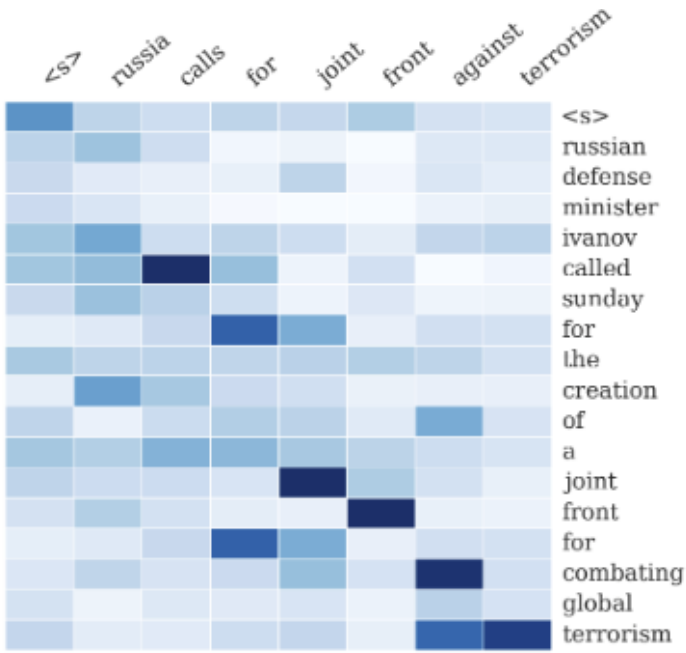


Figure 1: Example output of the attention-based summarization (ABS) system. The heatmap represents a soft alignment between the input (right) and the generated summary (top). The columns represent the distribution over the input after generating each word.

Figure 2.4.1

The attention mechanism allows the model to learn to focus on the most informative parts of the input sentence, which leads to more coherent and accurate summaries.

Additionally, the authors showed that the model is robust and can handle both long and short sentences with high accuracy. In that sense, the attention mechanism provides a more powerful and dynamic approach to sentence summarization, which can adapt to the specific characteristics of the input text.

In conclusion, the paper presents a novel neural attention-based model for sentence summarization that achieves state-of-the-art performance on several datasets. The authors demonstrate that the attention mechanism helps the model to focus on the most informative parts of the input sentence, which leads to more coherent and accurate summaries.

2.5 Attention is all you need

Authors: Ashish Vaswani

Year Of Publication:2017

Recurrent neural networks (RNN), long short-term memory networks(LSTM) and gated RNNs are the popular approaches used for Sequence Modelling tasks such as machine translation and language modeling. However, RNN/CNN handle sequences word-by-word in a sequential fashion. This sequentiality is an obstacle toward parallelization of the process. Moreover, when such sequences are too long, the model is prone to forgetting the content of distant positions in sequence or mix it with following positions' content.

Recent works have achieved significant improvements in computational efficiency and model performance through factorization tricks and conditional computation. But they are not enough to eliminate the fundamental constraint of sequential computation.

Attention mechanisms are one of the solutions to overcome the problem of model forgetting. This is because they allow dependency modeling without considering their distance in the input or output sequences. Due to this feature, they have become an integral part of sequence modeling and transduction models. However, in most cases attention mechanisms are used in conjunction with a recurrent network.

This paper introduces Transformer, a model architecture relying entirely on an attention mechanism to draw global dependencies between input and output.

The Transformer proposed in this paper is a model architecture which relies entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and tremendously improves translation quality after being trained for as little as twelve hours on eight P100 GPUs.

Neural sequence transduction models generally have an encoder-decoder structure. The encoder maps an input sequence of symbol representations to a sequence of continuous representations. The decoder then generates an output sequence of symbols, one element at a time.

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder.

The authors are motivated to use self-attention because of three criteria.

- One is the total computational complexity per layer.
- Another is the amount of computation that can be parallelized, as measured by the minimum number of sequential operations required.
- The third is the path length between long-range dependencies in the network.

The Transformer uses two different types of attention functions:

Scaled Dot-Product Attention, computes the attention function on a set of queries simultaneously, packed together into a matrix.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

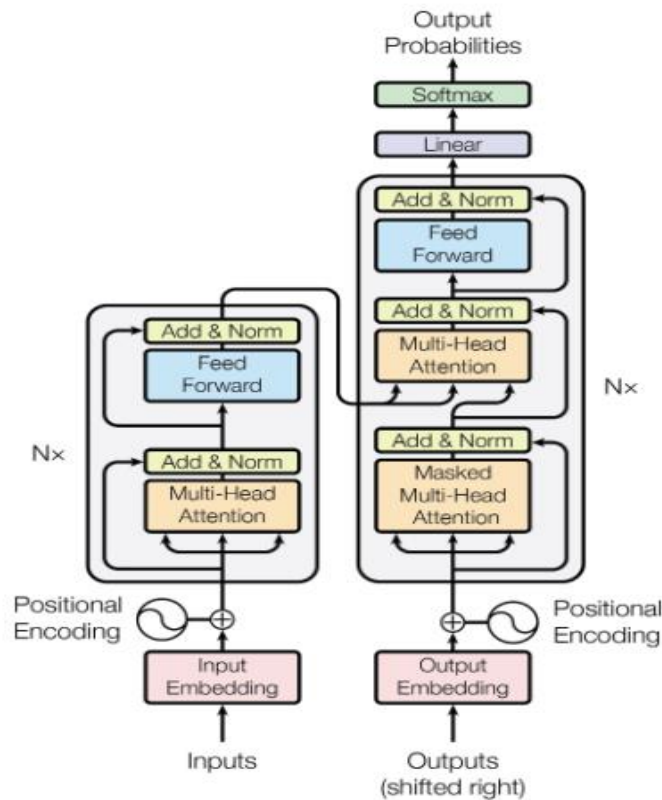


Figure 2.5.1

A self-attention layer connects all positions with a constant number of sequentially executed operations, whereas a recurrent layer requires $O(n)$ sequential operations.

In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence length is smaller than the representation dimensionality, which is often the case with machine translations.

Transformer has only been applied to transduction model tasks as of yet. In the near future, the authors plan to use it for other problems involving input and output modalities other than text. They plan to apply attention mechanisms to efficiently handle large inputs and outputs such as images, audio and video.

The Transformer architecture from this paper has gained major traction since its release because of major improvements in translation quality and other NLP tasks.

2.6 BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Authors: Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova

Year Of Publication: 2018

The BERT (Bidirectional Encoder Representations from Transformers) paper, introduced by researchers at Google AI, presents a breakthrough in natural language processing (NLP). BERT is a pre-training technique that utilizes a bidirectional Transformer model to generate context-rich word representations. Unlike previous approaches that relied on unidirectional models, BERT captures the bidirectional context of words, allowing it to better understand the meaning and relationships within sentences. BERT is trained on large amounts of unlabeled text, using masked language modeling and next sentence prediction tasks.

The key innovation of BERT lies in its pre-training and fine-tuning process. During pre-training, BERT learns to predict missing words within sentences by masking certain words and using the surrounding context. Additionally, BERT learns to predict whether two sentences appear consecutively in a document. This process enables BERT to acquire a deep understanding of syntax, semantics, and discourse structure. After pre-training, BERT is fine-tuned on specific downstream tasks, such as sentiment analysis or question answering, by training the model on labeled data. The pre-training and fine-tuning stages together make BERT a highly versatile and powerful language model.

BERT achieves remarkable results across a wide range of NLP tasks, surpassing previous state-of-the-art models. It demonstrates superior performance in tasks such as named entity recognition, part-of-speech tagging, and question answering. BERT's success can be attributed to its ability to capture both local and global contextual information, allowing it to understand complex sentence structures and make accurate predictions. The model's bidirectional nature and attention mechanism enable it to effectively handle long-range dependencies and improve the representation of words in context.

The impact of BERT extends beyond its performance on individual tasks. It has spurred advancements in various areas of NLP research, including transfer learning and contextual word representations. BERT's pre-training and fine-tuning methodology has inspired the development of numerous models that have achieved state-of-the-art

results in a wide range of applications. The paper's release has also fostered the creation of large-scale pre-trained language models, leading to significant progress in NLP. BERT's influence has been felt across academia and industry, driving innovation and enabling new possibilities for natural language understanding and generation.

System	Dev F1	Test F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	93.1
Fine-tuning approach		
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
Feature-based approach (BERT _{BASE})		
Embeddings	91.0	-
Second-to-Last Hidden	95.6	-
Last Hidden	94.9	-
Weighted Sum Last Four Hidden	95.9	-
Concat Last Four Hidden	96.1	-
Weighted Sum All 12 Layers	95.5	-

Table 2.6.1

Chapter 3

REQUIREMENT SPECIFICATION

3.1 FUNCTIONAL REQUIREMENTS

- Data Input and Preprocessing
 - Accept input in the form of a medical prescription image.
 - Preprocess input data to remove stop words, common words and whitespaces.
- Custom Dataset Generation
 - Creation of a small custom annotated dataset with Indian drug names.
- User Interface
 - User friendly interface where the user can select the trained model and upload the image of the prescription.
 - After the image is processed and the drug names are identified, it should give an output with a hyperlink to more information on the drug.
- Implement a custom Named Entity Recognition (NER) model to identify medical terms and medicines from the input data

3.2 NON-FUNCTIONAL REQUIREMENTS

System should have high accuracy in identifying medical terms and chemicals with minimal false positives/negatives

- System should have low response time for processing input and generating output images
- System should be reliable and available for use 24/7 with minimal downtime
- Ensure data privacy and security by handling medical prescription data in compliance with relevant regulations (e.g., HIPAA)
- System should be scalable to handle increasing volumes of medical prescriptions as the application grows

3.3 SYSTEM SPECIFICATIONS

3.3.1 Hardware Requirements

- Minimum CPU: Intel Core i5 or equivalent
- Minimum RAM: 8 GB
- Minimum storage: 256 GB SSD
- Recommended GPU: NVIDIA GeForce GTX 1060 or equivalent for accelerated NLP processing
- Internet connection with sufficient bandwidth for data retrieval and processing

3.3.2 Software Requirements

- Operating System: Windows 10 or Linux Ubuntu 18.04
- Python 3.7 or higher for software development and deployment
- Libraries and APIs: NLTK, SpaCy, TensorFlow, Keras for NLP processing, and image generation
- Integrated Development Environment (IDE): PyCharm or equivalent
- Version Control System: Git for code versioning and collaboration

4.1 WORKFLOW DIAGRAM

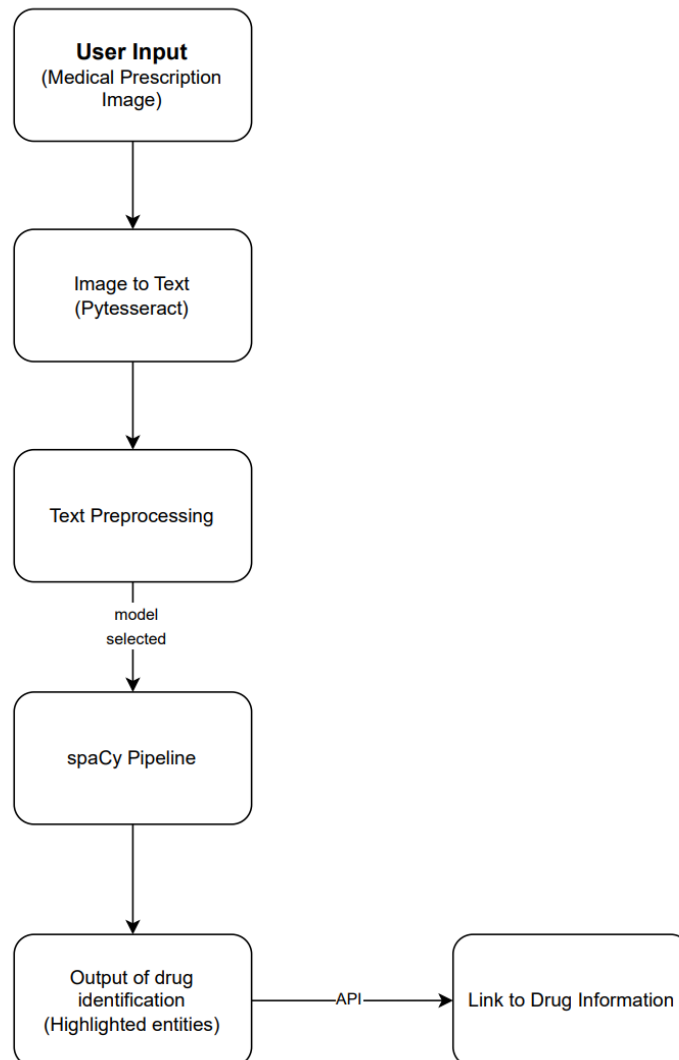


Figure 4.1.1

1. User gives the input - a medical prescription image
2. We pass the image through Pytesseract to perform OCR and convert the image to text.
3. Text preprocessing is performed where all the stop words, whitespaces and unwanted text is removed.
4. Preprocessed text is taken as input to the selected model and passed through the spaCy pipeline.
5. Output is generated where the drug names are the highlighted entities.

6. Identified drug entities are taken as input to an API to fetch drug related information from drugs.com.

4.2 SEQUENCE DIAGRAM

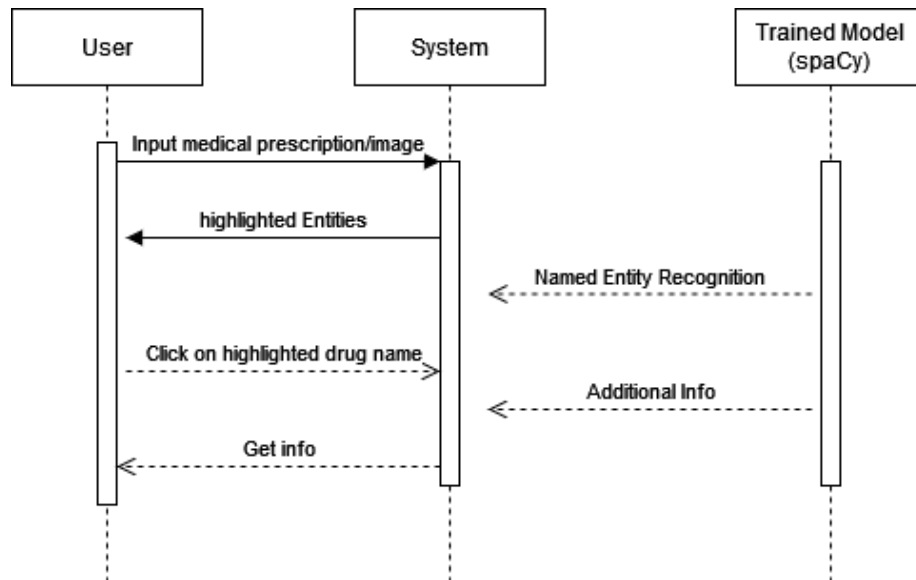


Figure 4.2.1

4.3 SPACY ARCHITECTURE

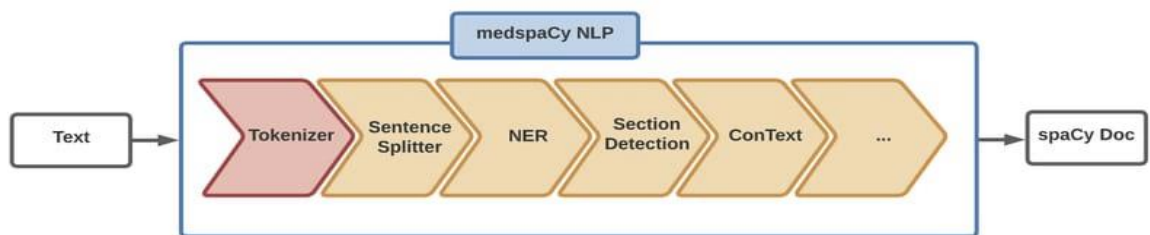


Figure 4.3.1

5.1 METHODOLOGY

5.1.1 Data Collection

The initial step in our project is to gather data from various sources. The data should be diverse and representative of the domain we are targeting.

1. **Data sources:** Identify potential sources from which prescription data can be collected. We have identified three suitable datasets for this project.

BC5CDR Dataset:

- The BioCreative V Chemical Disease Relation (BC5CDR) corpus is a manually annotated dataset for biomedical named entity recognition and relation extraction tasks. It consists of a collection of PubMed abstracts and full-text articles that have been annotated with chemical and disease entity mentions, as well as relations between them.
- The corpus is split into two parts: a training set and a test set. The training set contains 1,500 PubMed abstracts, while the test set contains 500 PubMed abstracts. Each abstract in the corpus is annotated by at least two annotators.
- The annotations in the BC5CDR corpus are encoded using the standoff format, which means that each entity and relation is assigned a unique identifier and stored separately from the original text. This format makes it easy to compare different annotations of the same text and to evaluate machine learning models on the corpus.
- The BC5CDR corpus has been used to evaluate a number of state-of-the-art machine learning models for biomedical named entity recognition and relation extraction tasks. It is a valuable resource for researchers in the biomedical natural language processing (NLP) community and has helped to advance the state of the art in this field.
- `en_ner_bc5cdr_md` is a pre-trained spaCy model for biomedical named entity recognition (NER) using the BioCreative V Chemical Disease Relation

(BC5CDR) dataset. This model is trained to recognize entities related to chemicals and diseases in biomedical text.

Here are some statistics about the corpus:

1. The corpus contains a total of 2,000 PubMed abstracts, which have been split into a training set (1,500 abstracts) and a test set (500 abstracts).
2. Each abstract in the corpus has been annotated by at least two annotators, and the annotations have been cross-checked to ensure accuracy.
3. The annotations in the corpus include chemical and disease entity mentions, as well as relations between them.
4. The annotations are encoded using the standoff format, which means that each entity and relation is assigned a unique identifier and stored separately from the original text.
5. The corpus is intended to be used for evaluating machine learning models for biomedical named entity recognition and relation extraction tasks.

It's important to note that the performance of machine learning models on the BC5CDR corpus can vary depending on a number of factors, including the specific task, the amount and quality of training data, and the evaluation metrics used.

[illegible]

Figure 5.1.1

Med7 Dataset:

- The Med7 dataset is a corpus of clinical text that has been annotated with named entities related to medication use. Here are some key characteristics of the Med7 dataset:
- The Med7 dataset contains a total of 5,000 clinical notes, which have been split into a training set (4,000 notes) and a test set (1,000 notes).
- The clinical notes are derived from the MIMIC-III database, which is a large, publicly available database of de-identified electronic health records.
- The annotations in the Med7 dataset include seven types of medication-related named entities: drug name, strength, dosage form, route of administration, frequency, duration, and reason for use.
- The Med7 dataset has been widely used for developing and evaluating machine learning models for medication-related named entity recognition in clinical text.
- The performance of machine learning models on the Med7 dataset can vary depending on the specific model architecture, training data, and evaluation metrics used.
- In a recent study, a transformer-based model achieved state-of-the-art performance on the Med7 dataset, achieving an F1 score of 0.903 on the test set.
- The Med7 dataset was introduced in a 2018 paper by Dernoncourt et al., titled "Training and Evaluating a Named Entity Recognizer for Electronic Health Records".
- The dataset is available in two formats: a raw text format, which includes the original clinical notes and corresponding annotations, and a tokenized format, which includes the text and annotations in a standardized format that is suitable for use with machine learning models.
- The Med7 dataset has been used in several studies that have explored different aspects of medication-related named entity recognition, including the use of different types of neural network architectures, the impact of pretraining on large corpora, and the use of active learning to improve model performance.

Label	Concept	Description
DOSAGE	1-2, sliding scale, taper, bolus, thirty (30) ml	The total amount of a drug administered
DRUG	aspirin, lisinopril, prednisone, vitamin b, flagyl	Generic or brand name of the medication
DURATION	for 3 days, 7 days, chronic, x5 days, for five more days	The length of time that the drug was prescribed for
FORM	tablet, capsule, solution, puff, adhesive patch, disk with device	A particular configuration of the drug which it is marketed for use
FREQUENCY	once a day, b.i.d., prn, q6h, hs, every six (6) hours as needed	The dosage regimen at which the medication should be administered
ROUTE	iv, p.o. (by mouth), gtt, nasal canula, injection,	The path by which the drug is taken into the body
STRENGTH	5mg, 100 unit/ml, 50mg/2ml, 0.05%, 25-50mg	The amount of drug in a given dosage

Figure 5.1.2

Custom Dataset:

The Custom Indian dataset is a collection of data obtained by scraping websites specific to the Indian context. Here are some key characteristics of the dataset :

- The Custom Indian Web Scraping dataset focuses on data collected from various websites in India. It aims to provide valuable information and insights related to a specific domain or research area relevant to the Indian context.
- The dataset is collected by utilizing web scraping techniques to extract relevant data from websites in India. The data collection process follows ethical considerations, respects website terms of service, and adheres to data protection regulations.
- Dataset captures information related to healthcare providers in India, including doctors, clinics, hospitals, and pharmacies. This aspect helps in analyzing prescription patterns, healthcare access, and provider-specific insights within the Indian healthcare landscape.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	text	labels													
2	The patient was prescribed Crocin for fever.	O O O B-DRUG O													
3	The doctor recommended Liv.52 for liver health.	O O B-DRUG O O													
4	The patient has been taking Ashwagandha for stress management.	O O O B-DRUG O O													
5	The Ayurvedic medicine Triphala is known for its digestive benefits.	O B-DRUG O O O O O													
6	The patient was diagnosed with diabetes and prescribed Diabecon.	O O O O O B-DISEASE O O B-DRUG													
7	The patient was prescribed Paracetamol for pain relief.	O O O B-DRUG O O													
8	The doctor recommended Ibuprofen for inflammation.	O O B-DRUG O													
9	The patient has been taking Aspirin for heart health.	O O O B-DRUG O O													
10	The pharmacist dispensed Amoxicillin for the patient's infection.	O O O B-DRUG O O O O													
11	The patient was advised to take Omeprazole for acid reflux.	O O O B-DRUG O O O													
12	The doctor prescribed Simvastatin for cholesterol management.	O O B-DRUG O O O													
13	The patient is allergic to Penicillin and should avoid it.	O O O B-DRUG O O O													
14	The medication Acetaminophen is commonly used for fever reduction.	O B-DRUG O O O O													
15	The patient was given Metformin for diabetes control.	O O O B-DRUG O O													
16	The doctor recommended Lisinopril for blood pressure regulation.	O O B-DRUG O O O O													
17	The patient is taking Atorvastatin for lipid control.	O O O B-DRUG O O													
18	The pharmacist dispensed Ciprofloxacin for the patient's urinary tract infectio	O O O B-DRUG O O O O													
19	The doctor prescribed Albuterol for asthma management.	O O B-DRUG O O O													
20	The patient has been using Hydrocortisone cream for skin irritation.	O O O B-DRUG O O O													
21	The medication Loratadine is commonly used for allergy relief.	O B-DRUG O O O O													
22	The patient was advised to take Warfarin for blood thinning.	O O O B-DRUG O O													
23	The doctor recommended Metoprolol for heart rate control.	O O B-DRUG O O O													
24	The patient is allergic to Sulfa drugs and should avoid them.	O O O B-DRUG O O O													
25	The pharmacist dispensed Fluoxetine for the patient's depression.	O O B-DRUG O O O													
26	The medication Pantoprazole is commonly used for gastric acid suppression.	O B-DRUG O O O O													
27	The patient was given Levothyroxine for thyroid hormone replacement.	O O O B-DRUG O O													
28	The doctor prescribed Sertraline for anxiety management.	O O B-DRUG O O O													
29	The patient has been using Mupirocin ointment for skin infection.	O O O B-DRUG O O O O													
30	The medication Gabapentin is commonly used for neuropathic pain.	O B-DRUG O O O O													
31	The patient was advised to take Ramipril for hypertension control.	O O O B-DRUG O O O													
32	The patient was diagnosed with hypertension and prescribed Amlodipine for	O O B-DISEASE O O B-DRUG O O O O O													
33	The doctor recommended Metronidazole for the treatment of bacterial infect	O O B-DRUG O O O O O B-DISEASE													
34	The patient has been using Prozac for the management of depression.	O O O B-DRUG O O B-DISEASE													

Figure 5.1.3

2. **Data Diversity:** Ensure that the collected data represent a diverse range of prescription types, including different medical conditions, medication classes, and patient demographics. This diversity helps train the automated system to handle various prescription formats, abbreviations, and language variations.
3. **Annotated Data:** Annotate the collected prescription data with the relevant entities to create a labeled dataset. The annotations may include medication names, dosages, instructions, patient details, and other pertinent information. Annotation can be performed by domain experts or trained annotators who are familiar with prescription structures and terminology.

When it comes to annotation, especially in natural language processing (NLP) tasks like named entity recognition (NER), the IOB (Inside-Outside-Beginning) tagging scheme is commonly used. IOB tagging allows annotators to label entities in text, indicating the position of each token within an entity. Here are some types of annotation commonly used with respect to IOB tagging.

Named Entity Recognition (NER) Annotation:

NER annotation involves identifying and labeling specific named entities in text, such as persons, organizations, locations, dates, or any other predefined entity types. Using IOB tagging, annotators assign labels to each token, indicating whether it is inside an entity (I), at the beginning of an entity (B), or outside an entity (O).

IOB Tagging:

- IOB tagging is a labeling scheme used to annotate named entities, chunks, relations, events, or semantic roles in natural language text.
- The acronym "IOB" stands for Inside, Outside, and Beginning, representing the three possible tags assigned to each token in the text.
- Each token is labeled with an IOB tag indicating its position relative to an entity or chunk. The three possible tags are "B" (beginning), "I" (inside), and "O" (outside).
- The "B" tag is assigned to the first token of an entity or chunk, indicating the beginning of a new entity or chunk.
- The "I" tag is assigned to tokens that are inside an entity or chunk, occurring after the beginning token.

- The "O" tag is assigned to tokens that are outside any entity or chunk, indicating that they do not belong to any labeled entity or chunk.
- IOB tagging allows for the representation of multiple entities or chunks within a single sentence by transitioning between "B" and "I" tags.
- The IOB tagging scheme enables the precise identification and differentiation of entities or chunks, facilitating subsequent processing or analysis tasks.
- IOB tags can be extended with additional labels or attributes to represent more complex structures or relations, depending on the specific task requirements.
- IOB tagging is widely used in various NLP tasks, and it serves as a standard for representing entity boundaries, event mentions, semantic roles, and other relevant linguistic structures within text data.

5.1.2 Data Preprocessing

Once the data is collected, we need to preprocess it to remove irrelevant information and clean it for further analysis. This may involve removing stop words, common words, white spaces, and any other unnecessary characters or symbols. This step ensures that our data is clean and ready for NLP processing.

5.1.3 NLP Pipeline

Next, we build a blank NLP pipeline that will consist of various processing steps such as tokenization, part-of-speech tagging and dependency parsing. These steps will be carried out using libraries such as NLTK, Spacy, or other relevant NLP libraries.

1. **Text Preprocessing:** The first step is to preprocess the prescription text to remove noise, normalize the text, and handle any special characters or formatting issues. This may involve tasks such as lowercasing the text, removing punctuation, and handling whitespace.
2. **Tokenization:** The prescription text is divided into individual tokens, such as words or sub words, using tokenization techniques. Tokenization allows the system to process the text on a granular level, facilitating subsequent analysis and extraction.
3. **Part-of-Speech (POS) Tagging:** POS tagging assigns grammatical labels (such as noun, verb, adjective) to each token in the prescription text. POS tags provide

information about the role and syntactic category of words, aiding in subsequent analysis and extraction steps.

4. **Named Entity Recognition (NER):** NER is a critical step in the pipeline. It identifies and extracts specific entities from the prescription text, such as medication names, dosages, instructions, and patient details. NER models are trained to recognize predefined entity types and label them accordingly.
5. **Entity Normalization:** After identifying entities, normalization techniques can be applied to standardize the extracted information. For example, medication names may be mapped to a standard drug database, dosages may be converted to a consistent format, and patient details may be validated against known naming conventions.
6. **Post-processing and Validation:** After the extraction of entities, post-processing steps are performed to validate and refine the extracted information. This may involve consistency checks, rule-based validation, or additional external validation sources to ensure accuracy and integrity of the extracted prescription details.

5.1.4 User Input Processing

When a user provides input to our system, we will process it by tokenizing the input text, tagging named entities and applying other NLP processing steps from our pipeline. This will allow us to extract relevant information from the user input.

5.1.5 Passing Entities To API

Once the highlighted entities are received, they are passed to the API to drugs.com which then searches its database to find the particular drug in question. If it exists, it returns back the user a clickable link that the user can visit to view more information.

5.1.6 Option to use custom dataset

Users have an option to choose between three datasets we have developed. Users can try out our custom dataset for Indian Medications to compare and review results.

5.2 Application Design

1. **User Interface:** Design an intuitive and user-friendly interface that allows users to input medical prescriptions. The interface should be visually appealing and easy to navigate, with clear instructions for users on how to input the prescription.
2. **Prescription Input Processing:** Develop a module that can take in the medical prescription as input and preprocess it to remove any irrelevant information, such as patient names or addresses. This module should also handle different formats of prescriptions, such as scanned images or digital text files, and convert them into a standardized format for further processing.
3. **NLP Pipeline:** Build a comprehensive NLP pipeline that includes steps such as tokenization, stop word removal, and entity recognition. This pipeline should be optimized for medical text and tailored to identify medical terms and chemicals involved in the prescription. This can be achieved by training a custom NER model on preprocessed data specific to the medical domain.
4. **Output Presentation:** Design a module that presents the output image along with the identified medical terms and chemicals in a meaningful way to the users. This can involve displaying the output image with annotated entities, providing a summary of the recognized entities, and offering options for further actions, such as saving the image or exporting the results.
5. **Error Handling:** Implement error handling mechanisms to handle any exceptions or errors that may occur during the processing of prescriptions. This can involve providing informative error messages to users, logging errors for troubleshooting, and implementing fallback options in case of failures.
6. **Deployment:** Develop a plan for deploying the application in a production environment, considering factors such as hosting options, server configuration, and system integration. Ensure that the deployment plan includes thorough testing and validation to ensure the system's stability and reliability in a production setting.

6.1 RESULTS

We will now see how the models discussed above have performed.

While evaluating a NER model there are usually 3 parameters that we generally see; Precision is a metric that assesses the accuracy of the NER system's predictions. It is defined as the ratio of true positive predictions (correctly identified named entities) to the total number of predicted named entities. In other words, precision measures the proportion of predicted named entities that are actually correct.

The precision score helps evaluate the system's ability to accurately identify and classify named entities without making many false positive errors. A higher precision score indicates that the system has a lower rate of false positives and is more precise in its predictions.

Precision can be calculated using the following formula:

$$\text{Precision} = (\text{True Positives}) / (\text{True Positives} + \text{False Positives})$$

Where:

True Positives (TP) are the number of correctly predicted named entities.

False Positives (FP) are the number of incorrectly predicted named entities.

It's important to note that precision should be considered alongside other evaluation metrics like recall and F1 score to get a comprehensive understanding of the performance of the NER system. Precision alone may not provide a complete picture, as a system with high precision but low recall may miss many actual named entities.

Recall is a metric that assesses the comprehensiveness or completeness of the NER system's predictions. It is defined as the ratio of true positive predictions (correctly identified named entities) to the total number of actual named entities present in the text. In other words, recall measures the proportion of actual named entities that were correctly identified by the system.

The recall score helps evaluate the system's ability to capture and detect named entities effectively without missing many of them. A higher recall score indicates that the system has a lower rate of false negatives and is more comprehensive in identifying named entities.

Recall can be calculated using the following formula:

$$\text{Recall} = (\text{True Positives}) / (\text{True Positives} + \text{False Negatives})$$

Where:

True Positives (TP) are the number of correctly predicted named entities.

False Negatives (FN) are the number of actual named entities that were not predicted by the system.

It's important to consider recall alongside other evaluation metrics like precision and F1 score to obtain a comprehensive understanding of the NER system's performance. Recall alone may not provide a complete picture, as a system with high recall but low precision may have a high number of false positives.

NER F-score, also known as the F1 score, is a commonly used evaluation metric in Named Entity Recognition (NER) tasks. It combines the precision and recall metrics into a single value that provides a balanced measure of the NER system's performance. The F1 score is the harmonic mean of precision and recall, calculated using the following formula:

$$\text{F1 Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

	Precision	Recall	F1
Med7	0.865	0.889	0.887
BC5CDR	0.884	0.873	0.884
CUSTOM	0.655	0.697	0.675

Table 6.1

6.2 PROJECT EXECUTION

6.2.1 Input and User Interface

A simple web-based interface where the user can choose the desired dataset (language model) and can upload an image of the medical prescription from the local system.

Select Language Model

Med7

Medical Prescription NER

Upload an image of a medical prescription to extract entities.

Upload an image

Drag and drop file here
Limit 200MB per file • PNG, JPG, JPEG

Browse files

JohnSmith-Example.jpg 24.6KB

Patient Name: John Smith

Address: 400 E 3rd Street
Duluth, MN 55804

Rx Amoxicillin 250 mg tablets # 42
TT tablets p.o. T.i.d. X 7 days

Figure 6.2.1

6.2.2 Output

Extracted Entities:

	Drug	Info
0	Amoxicillin	https://www.drugs.com/Amoxicillin.html

Clickable Links

[Amoxicillin](https://www.drugs.com/Amoxicillin.html)

Figure 6.2.2

The screenshot shows the Drugs.com website for Amoxicillin. The header includes the Drugs.com logo, a search bar, and navigation links. The main content area features the Amoxicillin title, pronunciation, generic name, brand names, and drug class. A sidebar on the right lists related topics like 'Antibiotics 101' and 'Drug Status'. The bottom section includes 'What is amoxicillin?' and 'User Reviews & Ratings'.

Drugs.com
Know more. Be sure.

Search [] All [] [] Register Sign In

Browse all medications: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z 0-9 Advanced Search

DRUGS A-Z | PILL IDENTIFIER | INTERACTIONS CHECKER | NEWS & ALERTS | PRO EDITION | MORE

Home > Amoxicillin [] Print [] Save

Amoxicillin []

Pronunciation: *am-ox-i-sil-in*
Generic name: amoxicillin
Brand names: Amoxil, Trimox, Moxatag
Drug class: Aminopenicillins

Medically reviewed by Kaci Durbin, MD. Last updated on Jan 3, 2022.

Uses | Warnings | Before taking | Side effects | Interactions | Dosage | FAQ

What is amoxicillin?

Amoxicillin is a penicillin antibiotic that fights bacteria. Amoxicillin is used to treat many different types of infection caused by bacteria, such as tonsillitis, bronchitis, pneumonia, and infections of the ear, nose, throat, skin, or urinary tract.

Amoxicillin is also sometimes used together with another antibiotic called clarithromycin (Biaxin) to treat stomach ulcers caused by Helicobacter pylori infection. This combination is sometimes used with a stomach acid reducer called lansoprazole (Prevacid).

There are many brands and forms of amoxicillin available and not all brands are listed on this leaflet.

Antibiotics 101

- List of Common Antibiotics & Types
- Antibiotics and Alcohol
- Probiotics and Antibiotics
- Antibiotics for Strep Throat
- Antibiotics for a Tooth Infection
- Antibiotics for UTIs

DRUG STATUS

- Rx** Availability: Prescription only
- Pregnancy & Lactation**: Risk data available
- CSA Schedule***: Not a controlled drug
- Approval History**: Drug history at FDA

User Reviews & Ratings

6.3 / 10 [] 359 Reviews

Images

Figure 6.2.3

7.1 CONCLUSION

The project on automated identification of medicine prescriptions has successfully addressed the need for an efficient and accurate system to extract relevant information from prescription texts. By leveraging natural language processing (NLP) techniques and custom named entity recognition (NER), the project has demonstrated the potential for automation and streamlining of the prescription identification process.

Throughout the project, we have explored various components of the NLP pipeline, including text preprocessing, tokenization, part-of-speech tagging, named entity recognition, entity normalization, dependency parsing, contextual understanding, and post-processing. Each of these stages has played a crucial role in accurately identifying and extracting key entities such as medication names, dosages, instructions, and patient details.

By collecting a diverse and well-annotated dataset, we were able to train a custom NER model that showcased promising performance in identifying prescription entities. The model was fine-tuned and evaluated to optimize its accuracy and generalizability. The iterative refinement process and continuous integration of user feedback have further improved the system's effectiveness in handling a wide range of prescription patterns and variations.

The automated identification of medicine prescription project has made significant strides in developing a robust and effective system for extracting prescription information. By leveraging the power of NLP techniques, custom NER models, and iterative refinement, the project has laid the foundation for improving the efficiency, accuracy, and reliability of prescription processing in healthcare settings. This project opens up new possibilities for automating tasks, reducing errors, and ultimately improving patient care in the field of medicine prescription management.

7.2 FUTURE SCOPE

The automated identification of medicine prescription project holds immense potential for future advancements and expansions. Here are some key areas of future scope for further development and improvement:

1. **Custom Dataset for Indian Medicine :** Generating a custom data for Indian drugs holds significant potential for future research and applications. By creating a comprehensive and localized dataset, several benefits can be realized :
 - a. Enhanced accuracy and relevance
 - b. Improved Drug identification
 - c. Easy to integrate with Indian use cases
2. **Enhanced Entity Recognition:** Continuously refining and expanding the entity recognition capabilities can lead to more accurate identification of medication names, dosages, instructions, and patient details. Incorporating advanced deep learning techniques, such as transformer-based models or domain-specific embeddings, can improve the system's ability to handle complex prescription patterns and variations.
3. **Multilingual Support:** Expanding the system's capabilities to support multiple languages can enable its adoption in diverse healthcare settings worldwide. Developing language-specific models and incorporating translation techniques can facilitate automated identification of medicine prescriptions in different linguistic contexts.
4. **Handling Ambiguities and Contextual Understanding:** Enhancing the system's ability to handle ambiguities, context, and implicit information can improve accuracy and user experience. Implementing advanced contextual understanding techniques, such as coreference resolution and context-aware language models, can help disambiguate entities and provide more meaningful insights from prescription texts.
5. **Adapting to Evolving Prescription Practices:** The project can continuously adapt to changing prescription practices, new medications, and emerging trends. Regularly updating the system with new annotated data and monitoring real-world prescription patterns can ensure its relevance and effectiveness over time.
6. **Real-time Processing and Feedback:** Enabling real-time processing of prescription texts can streamline healthcare workflows and improve efficiency. Providing instant feedback to users, such as highlighting potential errors or missing information in the prescription, can assist healthcare professionals in ensuring accuracy and completeness.

7. **Privacy and Security Enhancements:** Continuously enhancing privacy and security measures is crucial to protect sensitive patient information. Incorporating encryption techniques, access controls, and compliance with data protection regulations can ensure the confidentiality and integrity of prescription data.
8. **Collaboration and Industry Adoption:** Collaborating with healthcare providers, institutions, and pharmaceutical companies can facilitate the adoption and integration of the automated identification system into existing healthcare systems. Partnering with industry stakeholders can lead to real-world implementation, user feedback, and further improvements based on practical use cases.
9. **Expanding Application Areas:** Exploring the application of automated identification of medicine prescription beyond healthcare settings, such as pharmaceutical research, clinical trials, and drug safety monitoring, can unlock new possibilities and avenues for the project.
10. **User-Friendly Interfaces:** Developing user-friendly interfaces, such as mobile applications or web-based platforms, can simplify the interaction between healthcare professionals and the automated identification system. Intuitive interfaces can enhance usability, accessibility, and adoption of the system in various healthcare settings.

In conclusion, the future scope of the automated identification of medicine prescription project lies in continuously improving accuracy, expanding language support, adapting to evolving practices, integrating with decision support systems, ensuring privacy and security, fostering collaboration, and exploring new application areas. By pursuing these avenues, the project can contribute to the advancement of medication management, patient safety, and healthcare efficiency.

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